Post RTG Fellowship Essay

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The post-RTG funding would support me to get a solid start on research in my first semester following my oral exam, which I am taking at the end of October. With the extra time aided by the fellowship, I hope to get at a research problem that will eventually become my thesis.

These past semesters I have focused on gaining foundational knowledge in algebraic geometry, as well as pursuing several topics to find my interests. In particular, I have sampled

- Syzygies and minimal resolutions
- Stability of vector bundles and complexes
- Semiorthogonal decompositions of derived categories of sheaves (and related quivers)
- The derived category of a quadric, by understanding the spinor bundles through the representation theory of $\mathfrak{so}(Q)$.
- Some moduli, especially the construction of the Hilbert scheme.

Going forwards, I plan to understand the relationship between *pencils* of quadrics and hyperelliptic curves, from a derived category perspective using "Homological Projective Duality". This will make good use of my background in derived categories and my focus on quadrics.

Then, there is another angle from which I want to study these intersections of quadrics (and isoptropic Grassmannians) that Aaron has been introducing to me. A theorem of Ramanan-Bhosle describes the intersection of 2 quadrics in $\mathbb{P}^5$ as a certain moduli space of vector bundles on a curve C of genus 2. Then, this moduli space interacts with certain real/complex representations of the fundamental group of C due to a theorem of Narasimhan-Seshadri.

We are optimistic that we can extend this sort of interaction between intersections of quadrics, moduli of vector bundles, and fundamental group representations to higher dimensions. This project excites me—a lot of machinery that I enjoy is all coming together.